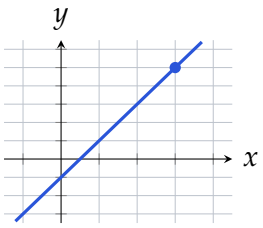
Lesson 1-2 • Limits to a Point of a Function

Notes (filled in)

Objective: evaluate limits algebraically, including when plugging in fails.

Evaluating limits

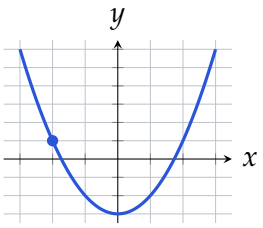
Linear $f(x) = 2x - 1$



$\lim_{x \to 3} f(x) =$ 5

$f(3) =$ 5

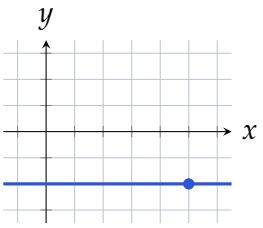
Quadratic $g(x) = x^2 - 3$



$\lim_{x \to -2} g(x) =$ 1

$g(-2) =$ 1

Constant $h(x) = -2$



$\lim_{x \to 5} h(x) =$ −2

$h(5) =$ −2

For polynomials (and constants), the limit equals the value: just plug in c.

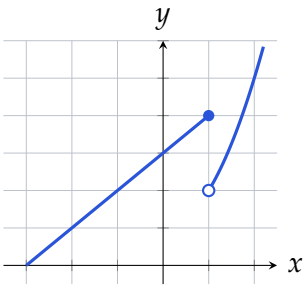
Limits of piecewise functions

$$j(x) = \begin{cases} x + 3, & x \leq 1 \\ x^2 + 1, & x > 1 \end{cases}$$

$x = 1$ is the suspect point : use the top piece for the left and the bottom piece for the right.

$\lim_{x \to 1^-} j(x) =$ 4 $\quad \lim_{x \to 1^+} j(x) =$ 2

$\lim_{x \to 1} j(x) =$ DNE $\quad j(1) =$ 4



When plugging in fails

$\lim_{x \to 3} \dfrac{x^2 - 9}{x^2 - x - 6}$ Plug in: $\dfrac{0}{0}$, the indeterminate form. It means: do more work!

$$\frac{(x-3)(x+3)}{(x-3)(x+2)} = \frac{x+3}{x+2} \to \frac{3+3}{3+2} = \frac{6}{5}$$

Three steps:

1) Factor
2) Cancel
3) Plug in

Example 1

$\lim_{x \to 0} \dfrac{x^2 - 4x}{3x^2 + x}$

$\frac{0}{0}$. $\dfrac{x(x-4)}{x(3x+1)} = \dfrac{x-4}{3x+1} \to \dfrac{-4}{1} = -4.$

Example 2	$\lim_{x\to -3} \dfrac{x^2+6x+9}{x+3}$ $\frac{0}{0}$. $\dfrac{(x+3)(x+3)}{x+3} = x+3 \to -3+3 = 0$. (A limit of 0 is a real answer, not DNE.)
Summary	Try plugging in first. A number means you are done; $\frac{0}{0}$ means factor, cancel, then plug in again. For a piecewise function, compare the left piece and the right piece at the suspect point.